

BANK OF LEEDS

Internal Investment Memorandum

Confidential – Infrastructure & Strategic Investments Division

Humber Hydrogen Superhub

A Strategic Energy & Industrial Investment Proposal

Integrating the Humber Barrage, Cleethorpes Nuclear Power Station and Britain's Largest Green Hydrogen Economy

Prepared for:

Investment Committee

Prepared by:

Infrastructure Strategy Unit

Status:

Concept Proposal for Deep Research Commission

Executive Summary

The Humber Estuary is already the UK's largest industrial cluster by emissions and hosts major ports, refineries, chemical plants, and offshore wind infrastructure. UK Government strategies have also identified the Humber as a leading location for hydrogen production and carbon capture. This memorandum considers a long-term scenario in which additional major infrastructure—specifically a tidal barrage and a new nuclear power station—is evaluated as part of a regional hydrogen strategy rather than as stand-alone projects.

This proposal investigates whether the Humber could become Europe's lowest-cost large-scale hydrogen production region through three complementary energy sources:

- Offshore wind
- A Humber tidal barrage (assumed annual output approximately **7.5 TWh**)
- A new nuclear generating station near Cleethorpes

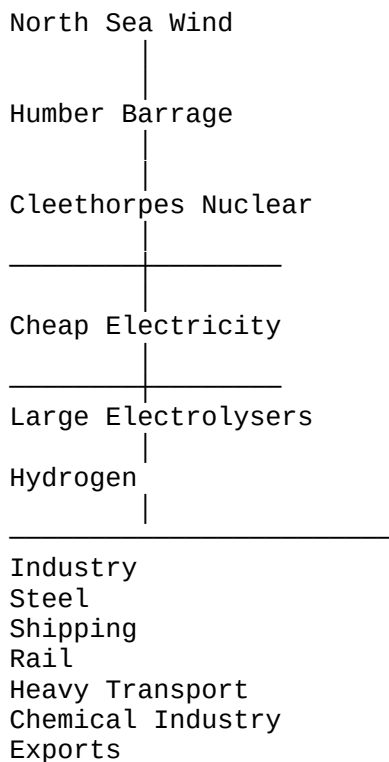
The objective would be to create an integrated **Hydrogen Superhub** supplying:

- British industry
- Heavy transport
- Aviation fuels
- Maritime fuels

- Export markets
 - Long-duration energy storage
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Strategic Vision

Rather than viewing electricity generation and hydrogen separately, the proposal considers an integrated regional energy system:



The central economic hypothesis is:

Hydrogen prices are primarily determined by electricity costs, electrolyser utilisation, financing costs, and distribution infrastructure.

Why the Humber?

The Humber possesses an unusual concentration of assets:

Existing Ports

- Hull
- Immingham
- Grimsby
- Goole

Industrial Users

- Refineries
- Chemicals
- Fertiliser
- Steel supply chains
- Glass
- Food processing

Offshore Wind

The Humber already supports one of the world's largest offshore wind servicing clusters.

Existing Hydrogen Plans

Government and industry have identified the Humber as a priority region for hydrogen development.

Proposed New Infrastructure

Phase One

Humber Barrage

Concept:

Annual generation

7.5 TWh

Potential benefits:

- Predictable tidal electricity
 - Flood protection (subject to engineering assessment)
 - Transport crossing opportunities
 - Grid stability
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Phase Two

Cleethorpes Nuclear Station

Concept:

One modern large reactor or equivalent advanced reactor capacity.

Purpose:

Provide:

- continuous baseload electricity
- high-capacity industrial power
- stable electrolysis supply

Unlike wind power, nuclear generation enables electrolyzers to operate at consistently high utilisation, potentially reducing the cost of hydrogen production by spreading capital costs across more operating hours.

Phase Three

Humber Hydrogen Valley

Construct Britain's largest electrolysis complex.

Potential scale:

Several gigawatts of electrolysis over multiple development phases.

Products:

- Hydrogen
 - Oxygen
 - Industrial heat
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Lowering Hydrogen Costs

Hydrogen economics are highly sensitive to electricity input costs and electrolyser utilisation.

Potential cost reductions could come from:

1. Very High Asset Utilisation

Electrolysers powered only by intermittent renewables often have lower annual operating hours.

Combining:

- nuclear
- tidal
- offshore wind

could increase utilisation, lowering fixed costs per kilogram of hydrogen, although the exact outcome depends on electricity pricing, dispatch strategy and financing.

2. Low-Cost Electricity

Hydrogen production is dominated by electricity costs.

Potential sources:

- Nuclear
- Wind
- Tidal

A diversified generation mix can reduce exposure to periods of high wholesale electricity prices.

3. Large Electrolysers

Economies of scale.

Example:

100 MW

↓

500 MW

↓

1 GW

↓

5 GW

Larger facilities generally reduce unit capital costs through scale and standardisation.

4. Shared Infrastructure

One hydrogen network instead of multiple small projects.

Shared:

- pipelines
 - compressors
 - storage
 - export terminals
 - grid connections
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5. Industrial Clustering

The Humber already contains many potential hydrogen users.

Minimal transport distance.

Minimal distribution cost.

6. Salt Cavern Storage

Investigate underground storage opportunities where suitable geology exists.

Benefits:

- seasonal storage
 - balancing intermittent generation
 - strategic reserve
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Potential Hydrogen Economy

Illustrative priority sectors include:

Sector	Potential Use
Steel	Direct reduced iron (where technically applicable)
Refineries	Process hydrogen
Fertiliser	Ammonia production
Shipping	Marine fuels (direct or derivative fuels)
Aviation	Synthetic fuels
Heavy trucks	Fuel cell applications where competitive
Rail	Non-electrified routes where appropriate
Export	Ammonia or liquid hydrogen (subject to market economics)

Infrastructure Programme

The proposal considers:

- Humber Barrage
 - Cleethorpes Nuclear Station
 - High-voltage transmission upgrades
 - Gigawatt-scale electrolyzers
 - Hydrogen pipeline network
 - Port export facilities
 - Hydrogen storage
 - Industrial conversion
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Economic Benefits

Potential impacts to be tested include:

Construction:

- major engineering employment over a decade or more

Operations:

- long-term skilled jobs in energy and industry

Industrial competitiveness:

- lower-carbon feedstocks
- increased energy security
- attraction of new manufacturing investment

Export potential:

- hydrogen and derivative fuels
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Role of the Bank of Leeds

The memorandum recommends evaluating a dedicated **Humber Energy Infrastructure Fund**.

Possible investment areas:

- Electrolyser facilities
- Hydrogen pipelines
- Industrial energy parks
- Storage infrastructure
- Export terminals
- Grid infrastructure
- District energy systems

Rather than owning generation assets directly, the Bank could focus on financing or owning regulated infrastructure with long-lived cash flows.

Financial Structure

Potential funding sources:

- UK Infrastructure Bank or successor public financing mechanisms
- Pension funds
- Insurance companies
- Infrastructure funds
- Green bonds
- Industrial customers through long-term offtake agreements

Illustrative financing would depend on detailed project appraisal, market conditions and government policy.

Deep Research Programme

The Bank of Leeds should commission a comprehensive study with the following workstreams:

Volume I – Humber Energy Resource Assessment

- Offshore wind integration
- Tidal barrage engineering and generation profile
- Nuclear siting and grid integration

Volume II – Hydrogen Cost Optimisation

- Electrolyser technologies
- Electricity sourcing strategies
- Water supply and treatment
- Storage technologies

Volume III – Industrial Demand

- Current and future hydrogen demand
- Industrial conversion pathways
- Export opportunities

Volume IV – Infrastructure

- Pipeline corridors
- Port upgrades
- Storage geology
- Grid reinforcement

Volume V – Financial Analysis

- Capital expenditure
- Operating expenditure
- Levelised cost of hydrogen under multiple scenarios
- Revenue modelling
- Public–private partnership options
- Sensitivity analysis

Volume VI – Environmental & Regulatory Review

- Planning requirements
- Marine impacts
- Water abstraction
- Biodiversity
- Carbon accounting

Recommendation

The Infrastructure Strategy Unit recommends commissioning a **250–400 page multidisciplinary feasibility study** titled:

"The Humber Hydrogen Superhub: Technical, Economic and Financial Feasibility Study for an Integrated Offshore Wind, Tidal and Nuclear Hydrogen Economy."

The study should determine whether combining renewable generation, potential tidal generation, and firm low-carbon electricity can materially reduce hydrogen production costs compared with standalone projects, while identifying the infrastructure, financing structures and industrial demand required to support a commercially viable regional hydrogen hub.